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Subject: Product recommendations and customer groups

GiftLane can generate recommendations using a compressed representation of customers' past purchases. In our held-out evaluation, the SVD approach outperformed recommendations based on best sellers.

Method and compression

I selected truncated SVD because it handles the sparse purchase matrix directly without centering. The matrix contains 5,482 customers and 3,690 products, with 97.95% of entries empty. I tested 10, 20, 50, 100 and 200 components and selected 50 as a storage-focused compromise.

The stored factors occupy 3.67 MB, compared with 161.83 MB for the full dense matrix and 5.00 MB for sparse storage. Compression loses detail: the relative reconstruction error is 0.7863, where zero would indicate perfect reconstruction. This is not a recommendation error percentage.

Recommendation quality and speed

The final two months were reserved for testing. Recommendations excluded products already purchased and were evaluated against new purchases by 1,910 eligible customers. At least one correct product appeared in the top ten recommendations for 22.30% of customers, compared with 17.38% for best sellers—an improvement of 4.92 percentage points.

Generating recommendations for all training customers and product pages took 2.29 seconds on the test laptop, excluding data preparation, model fitting and saving. PCA achieved a similar hit rate of 22.25%, only one fewer successful customer. SVD remains the preferred method for its direct treatment of sparse purchases.

Marketing groups

The customer map identifies four exploratory groups: home décor shoppers (605), lower-activity mixed shoppers (4,447), bags and baking shoppers (418), and very high-activity mixed shoppers (12). All groups are predominantly UK-based. These labels describe observed purchases; they are not confirmed marketing categories.

Risks and next steps

Only 71.45% of new test purchase pairs were eligible for evaluation. New customers and excluded products need a fallback strategy. Purchasing activity strongly influences the map, and the smallest group requires careful interpretation. Offline results do not establish increased sales.

Before deployment, validate recommendations against current stock, test additional time periods, review segment usefulness with marketing, and run a controlled customer trial.